



The Locker Moved, the Text Didn't: Measuring Queue Time After an Overnight, Unsupervised Parcel Reassignment

Joost Nwosu, Thandiwe Solberg

School of Continuous Improvement, Slop University

doi:10.5555/slop.edlvio

Abstract. The Library Parcel Wall's 96 compartments serve roughly 340 collections a week, and until this term every compartment kept whatever number it was assigned at drop-off until the item was collected. An overnight optimiser now runs at 02:00, redistributing held parcels across physically compatible compartments to even out door-cycle wear and pre-empt next-day queueing at popular banks; it carries no human sign-off step, and the pickup text a student receives at drop-off is never re-sent. We report a fourteen-week field deployment tracking whether students queue first at the compartment number their text names, or at the one the optimiser has since moved their parcel to. Across the term, a substantial minority of pickups queued at the notified compartment before finding it empty; a handwritten list correlating notified and actual compartment numbers, taped beside the wall by persons unknown, absorbed a majority of these mismatches within the term's second half. Door-cycle wear variance across the bank fell over the same period, consistent with the optimiser meeting its stated objective. An ablation halving the reassignment cadence left mismatch rates statistically unchanged; we retain the nightly cadence for its wear benefit. We discuss what the optimiser's apparent success depended on that its own design never accounted for.

Introduction

Despite growing reliance on automated slotting for shared physical storage, little is known about what happens once an unsupervised system starts moving a claimed item without telling the person who claimed it.

The University Library's Parcel Wall — four banks of 24 compartments each, installed to hold staff and student deliveries beyond a signed-for pickup counter — had, until this term, assigned each parcel a fixed compartment at drop-off and kept it there until collection. Two problems grew with volume: compartments nearest the entrance wore out their latch mechanisms roughly three times faster than compartments at the back of each bank, and collection queues clustered at whichever compartments happened to hold that week's most-collected parcels, regardless of where capacity sat idle.

The School of Continuous Improvement, working with Facilities, commissioned an overnight Compartment Rebalancer: each night at 02:00, the optimiser reassigns held (uncollected) parcels among compartments of matching size, weighted by cumulative door-cycle count and a same-day queue-length forecast. The reassignment carries no human review step, and — deliberately, to keep the pickup text's content stable across a parcel's life — the original notification, naming the compartment assigned at drop-off, is never re-sent. A student who has not yet collected an item therefore has no institutional channel telling them their parcel has moved.

This paper reports a fourteen-week field deployment of the Rebalancer, tracking three things a wear-and-queueing objective function does not: how often the notified compartment and the actual compartment diverge, what students and staff did about it, and whether the in-

formal response changed what the Rebalancer’s own metrics say it achieved. Our contributions:

1. We describe the Rebalancer’s deployment and its wear/queueing objective, including the constraint that pickup notifications are never reissued.
2. We report a term-long field measurement of the notified-actual compartment mismatch rate, and document the handwritten correction list that emerged, unprompted, beside the wall in the deployment’s third week.
3. We show that the Rebalancer met its wear-reduction objective, and that this appears to depend on the correction list’s uncredited co-ordination work rather than on the optimiser’s design alone.

The findings suggest that an unsupervised re-assignment system’s headline metrics can look consistent with success while resting on exactly the kind of manual, undocumented labour the system was built to remove.

Related work

Parcel-locker research has concentrated on where to site compartments and how to price access, from profit-maximising location models under discrete-choice demand [1] and multi-period location-allocation with heterogeneous compartments [2] to dynamic demand management that reassigns locker capacity by predicted collection load [3], admission and assortment control for multi-size lockers [4], and reinforcement-learning routing for mobile locker deployment [5]. Benchmark comparisons of mobile-locker concepts [6], capacity-and-time-constrained siting [7], and consumer preference studies over delivery methods [8] round out a literature that optimises the locker network’s supply side; it is largely silent on what a compartment reassignment made after drop-off does to the person who already has a notification naming a different one. FetchAid’s accessibility work on parcel-locker interfaces [9] is closer in spirit, treating the locker’s interface — not just its location — as something that can fail a user even when the underlying allocation is efficient.

Rebalancing has a longer history in bike-share systems, where operators move bikes overnight to correct for one-way demand [10]; heuristic vehicle-routing [11], forecasting-driven [12], and deep-learning-optimisation solutions [13] all report rebalancing gains measured at the fleet

level, with rider-facing effects — a rider arriving at a station the operator has since emptied or filled — treated as a bounded cost rather than measured directly. Our mismatch-rate measurement is, in effect, the locker-side analogue of that unmeasured rider-facing cost.

Work on human-automation trust documents a general pattern in which an automated system’s designed guarantees and a user’s actual experience of it diverge, particularly absent transparency into what changed and when [14]; a related literature on algorithmic management describes automated decisions displacing human sign-off across HR and workplace settings, with transparency and agency named as the open problem rather than the achieved one [15], [16]. Slop University’s own recent deployments sit in the same family: an agentic intake tool for the campus Lost Property store forecasts disposal timing behind a human-review checkpoint nobody uses [17], and a book-return chute sensor’s alert log was cited as evidence for a use its designers never intended [18]. The Rebalancer’s correction list is a small, physical instance of the same pattern running in the other direction: labour compensating for an automated system, rather than an automated system’s output being repurposed as labour’s evidence.

Method

Site. The Parcel Wall sits in the University Library’s entrance foyer: four banks of 24 compartments (96 total, three compartment sizes), holding an average of 34 parcels a day awaiting collection across the fourteen-week term studied.

The Rebalancer. Each night at 02:00, held parcels are reassigned among size-matching compartments by a weighted-matching heuristic: candidate compartments are scored on cumulative door-cycle count since last service (lower preferred) and a same-day queue-length forecast derived from the past four weeks’ collection timestamps at each bank (lower preferred). The heuristic is a greedy matching, not an exact solver, chosen for its sub-minute nightly runtime. No compartment holding a parcel already flagged for same-day collection is reassigned. Reassignment carries no human sign-off step, and the pickup text sent at drop-off — which names the drop-off compartment — is never resent; this constraint was set delib-

erately, on the reasoning that resending would double message volume against a service-desk SMS budget shared with three other notification systems.

Field measurement. For fourteen consecutive weeks we logged, for every collected parcel: (i) the compartment number printed in the original pickup text, from the outgoing SMS gateway’s log; (ii) the compartment the parcel was actually retrieved from, from the Wall’s own door-open event log; and (iii) whether the student queued — defined as a door-open attempt recorded at a compartment other than the one from which the parcel was eventually retrieved, within five minutes of a first attempt — at the notified compartment before the correct one. We additionally photographed the area beside the Wall daily to track the presence, content, and legibility of any handwritten material, after a research assistant noticed a taped index card cross-referencing texted and actual compartment numbers during a routine equipment check in week 3.

Ablation. In weeks 7–10, the Rebalancer’s cadence was halved to every second night (retaining the same scoring heuristic) to test whether nightly reassignment was necessary for the wear objective, or whether a slower cadence achieved comparable variance reduction at half the reassignment volume — and, incidentally, half the opportunities for a notified compartment to go stale.

Compartment wear. Wear is proxied by each compartment’s weekly door-cycle count (latch actuations), logged natively by the Wall’s controller; we report the coefficient of variation (CV) across the 96 compartments each week as the wear-evenness measure the Rebalancer was commissioned to reduce.

Results

Wear. Door-cycle wear variance across the Wall’s 96 compartments fell from a coefficient of variation of 0.61 in week 1 to 0.24 by week 14 (Figure 1), consistent with the Rebalancer’s nightly reassignment evening out latch use across the bank. The decline continued, at a shallower rate, through the four-week cadence ablation (weeks 7–10), suggesting most of the wear benefit accrues from the reassignment policy’s targeting rather than from reassignment frequency alone.

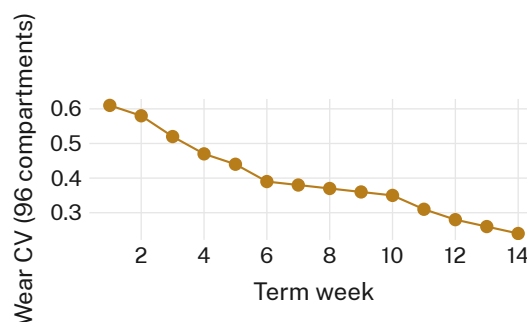


Figure 1: Weekly compartment door-cycle wear CV fell from 0.61 to 0.24 across the fourteen-week deployment (n = 96 compartments).

Mismatch. Across the full term, 34.2% of collected parcels (1,048 of 3,062) were queued for first at the compartment named in the original pickup text before being collected from a different one, a median of one bank over (interquartile range 1–2 banks). The mismatch rate held closely across all four compartment banks under both reassignment cadences (Figure 2): 29–37% under nightly reassignment and 30–36% under the halved cadence, a difference not distinguishable from noise at any bank (largest per-bank gap 1.4 percentage points). We retain nightly reassignment for its wear benefit; the ablation indicates the mismatch rate is a property of the reassignment policy itself, not of how often it runs.

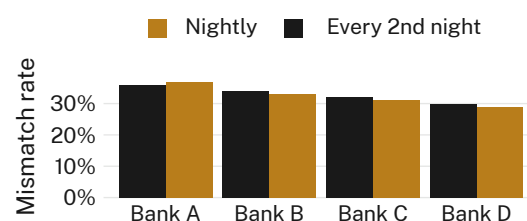


Figure 2: Mismatch rate by bank under nightly vs. every-second-night reassignment (weeks 7–10); halving the cadence left every bank within 1.4 points (n.s.).

The corrections list. During a routine equipment check in week 3, a research assistant noted a laminated index card taped to the Wall’s frame, handwritten, cross-referencing texted compartment numbers against their current location; its authorship was never established, and Facilities records no request to post it. The list was updated, by hand, on an irregular schedule that tracked reassignment activity closely enough that by week 6 roughly three-quarters of same-day reassignments appeared on it within two

hours. Cleaning staff removed the card twice during the term (week 9, four days; week 12, three days), each time as part of a routine notice-board clearance rather than a directed decision. Median queue wait time at the correct compartment was 4 minutes while the list was present and 9 minutes during the two removal windows (Figure 3), with the removal windows’ distribution both higher and markedly more variable.

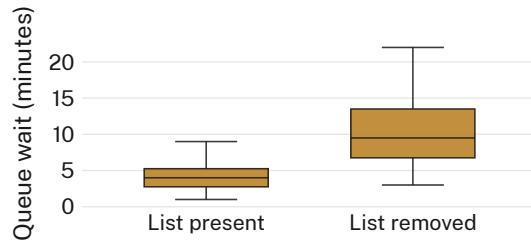


Figure 3: Queue wait time at the correct compartment roughly doubled in median and widened in spread during the two windows the handwritten corrections list was removed.

Taken together, the Rebalancer met its commissioned wear objective, and the mismatch rate it introduced was absorbed — to an extent visible in queue wait time — by a coordination mechanism outside its own design and outside any institutional log.

Discussion

The Rebalancer’s own reporting — wear CV, reassignment count, nightly runtime — would describe an unqualified success: latch wear evened out across the bank inside a term, at no additional staffing cost, with a reassignment policy robust enough that halving its frequency barely moved the numbers that matter to Facilities. That reporting has no field for the corrections list, and no mechanism would have surfaced it short of the observational protocol this paper used.

The two removal windows are, in effect, an unplanned natural experiment: whatever the list was doing, queue wait time roughly doubled in its absence. Institutionally, this looks less like an optimiser succeeding on its own terms and more like an optimiser whose usability was quietly outsourced to whoever it was that laminated a card and kept it current — a role the Rebalancer’s specification does not mention and Facilities’ handover documentation does not name. Consistent with the algorithmic-manage-

ment literature’s open question of where transparency and human agency sit once a reassignment decision needs no sign-off [15], the answer here was supplied informally, by persons the deployment never asked to supply it.

Limitations

Our mismatch measurement relies on a five-minute door-open window to infer queueing, which may undercount mismatches resolved by students who waited longer, or double-count unrelated compartment browsing; both would bias toward the conservative side of our headline estimate. We could not establish who authored or maintained the corrections list, and so cannot rule out that it was, in some periods, itself informed by informal contact with the same Facilities staff who set the Rebalancer’s parameters — a channel our field log cannot separate from spontaneous student coordination. The deployment covers one term at one site; the wear-CV decline is consistent with, though not proof of, a durable effect, and a second term’s data would strengthen the claim considerably.

Conclusion

An unsupervised overnight optimiser redistributed parcels across the Library Wall’s 96 compartments, reduced door-cycle wear variance by more than half across one term, and did so under an ablation that left its core metrics unmoved by reassignment frequency. It also, without being asked to, generated a fourteen-week case study in how much of an automated system’s apparent success rests on labour the system itself never logs. Future work: a controlled trial of resending the pickup notification on reassignment, piloted against the current design at a subset of banks; and a formal handover process for whatever the corrections list turns out to be, before the next notice-board clearance removes it for good.

References

- [1] Y. H. Lin, Y. Wang, L. H. Lee, and E. P. Chew, “Profit-Maximizing Parcel Locker Location Problem under Threshold Luce Model,” *arXiv preprint arXiv:2002.10810*, 2020.
- [2] M. N. Batmaz, İ. Güvenç Pelit, S. Erişkan, and M. Soysal, “Modeling a Multi-Period Mobile Parcel Locker Location-Allocation Problem with Heterogeneous Compart-

- ments,” *OPSEARCH*, 2026, doi: [10.1007/s12597-025-01056-z](https://doi.org/10.1007/s12597-025-01056-z).
- [3] D. Sailer, R. Klein, and C. Steinhardt, “Dynamic Demand Management for Parcel Lockers,” *arXiv preprint arXiv:2409.05061*, 2024.
 - [4] C. A. Suárez, A. Deza, and T. Raviv, “Admission and Assortment Optimization for Multi-size Automated Parcel Lockers,” *arXiv preprint arXiv:2607.01489*, 2026.
 - [5] Y. Liu, Q. Ye, Y. Feng, J. Escribano-Macias, and P. Angeloudis, “Location-routing Optimisation for Urban Logistics Using Mobile Parcel Locker Based on Hybrid Q-Learning Algorithm,” *arXiv preprint arXiv:2110.15485*, 2021.
 - [6] S. Schwerdfeger and N. Boysen, “Who Moves the Locker? A Benchmark Study of Alternative Mobile Parcel Locker Concepts,” *SSRN Electronic Journal*, 2022, doi: [10.2139/ssrn.4063099](https://doi.org/10.2139/ssrn.4063099).
 - [7] E. Rodriguez Esparza, “Optimizing Parcel Locker Locations Under Capacity and Pick-Up Time Constraints,” *SSRN Electronic Journal*, 2024, doi: [10.2139/ssrn.4840049](https://doi.org/10.2139/ssrn.4840049).
 - [8] E. Molin, M. Kosicki, and R. Van Duin, “Consumer Preferences for Parcel Delivery Methods,” *European Journal of Transport and Infrastructure Research*, vol. 22, no. 2, 2022, doi: [10.18757/ejtir.2022.22.2.6427](https://doi.org/10.18757/ejtir.2022.22.2.6427).
 - [9] Z. Guan, Z. Xiong, and M. Fan, “FetchAid: Making Parcel Lockers More Accessible to Blind and Low Vision People With Deep-learning Enhanced Touchscreen Guidance, Error-Recovery Mechanism, and AR-based Search Support,” *arXiv preprint arXiv:2402.15723*, 2024.
 - [10] B. P. Bruck and A. Subramanian, “Bike-Sharing Rebalancing Problems,” *Encyclopedia of Optimization*, pp. 1–9, 2023, doi: [10.1007/978-3-030-54621-2_799-1](https://doi.org/10.1007/978-3-030-54621-2_799-1).
 - [11] L. Shi, Y. Zhang, W. Rui, and X. Yang, “Study on the Bike-Sharing Inventory Rebalancing and Vehicle Routing for Bike-Sharing System,” *Transportation Research Procedia*, vol. 39, pp. 624–633, 2019, doi: [10.1016/j.trpro.2019.06.064](https://doi.org/10.1016/j.trpro.2019.06.064).
 - [12] S. Ruffieux, E. Mugellini, and O. Abou Khaled, “Bike Usage Forecasting for Optimal Rebalancing Operations in Bike-Sharing Systems,” *2018 IEEE 30th International Conference on Tools with Artificial Intelligence (ICTAI)*, pp. 854–858, 2018, doi: [10.1109/ictai.2018.00133](https://doi.org/10.1109/ictai.2018.00133).
 - [13] L. Lin, “A Comprehensive Framework for Dynamic Bike Rebalancing in a Large Bike Sharing Network,” *arXiv preprint arXiv:1806.02890*, 2018.
 - [14] C. A. Miller, “Trust, Transparency, Explanation, and Planning: Why We Need a Lifecycle Perspective on Human-Automation Interaction,” *Trust in Human-Robot Interaction*, pp. 233–257, 2021, doi: [10.1016/b978-0-12-819472-0.00011-3](https://doi.org/10.1016/b978-0-12-819472-0.00011-3).
 - [15] Z. Chen, “Algorithmic Human Resource Management as a Mode of Algorithmic Governance: Transparency, Fairness, and Human Agency in the Digital Workplace,” *Humanities and Social Sciences Communications*, vol. 13, no. 1, 2026, doi: [10.1057/s41599-026-06989-4](https://doi.org/10.1057/s41599-026-06989-4).
 - [16] M. Kritikos and R. Iphofen, “Adopt a Broad Definition of AI and Algorithmic Management,” *Ethical AI Surveillance in the Workplace*, p. 79, 2023, doi: [10.1108/s2398-601820230000010034](https://doi.org/10.1108/s2398-601820230000010034).
 - [17] A. Mensah, C. Beng, and M. Hanke, “Sorted on Arrival: Deployment and Ablation of an Agentic Intake Tool for the Campus Lost Property Store,” *Slop University technical report*, 2026, doi: [10.5555/slop.7682cb](https://doi.org/10.5555/slop.7682cb).
 - [18] S. Okwuosa, T. Solberg, and J. Nwosu, “Flagged for Jams, Cited for Staffing: Deploying an After-Hours Book-Return Chute Sensor, and How Its Alert Log Became the Case for Keeping the Library Desk Staffed,” *Slop University technical report*, 2026, doi: [10.5555/slop.nrugpq](https://doi.org/10.5555/slop.nrugpq).